

Healthcare Analytics Dashboard Project Using Tableau

The objective of this project is to help hospital management monitor patients, doctors, appointments, treatments, and revenue through interactive dashboards.

Initial Supporting KPIs

Create the following KPIs as the initial step:

- **Total Patients:** COUNTD(patient_id)
- **Total Revenue:** SUM(Amount)
- **Total Appointments:** COUNT(Appointment ID)
- **Total Doctors:** COUNTD(Doctor ID)

The hospital management aims to use data-driven decision-making to improve operational efficiency, patient care, and revenue performance by addressing the following business questions.

Executive Dashboard Questions

1. How many patients are being handled by the hospital?

KPI:

- Total Patients

Business Value:

- Understand hospital workload

2. What is the total revenue generated?

KPI:

- Total Revenue

Business Value:

- Monitor the hospital's financial performance

3. Are appointments increasing over time?

Visualization:

- Monthly Appointment Trend

Business Value:

- Identify growth patterns

Doctor Analytics Questions

4. Which doctors handle the highest number of patients?

Visualization:

- Doctor-wise Appointments

Business Value:

- Support resource allocation

5. Which specialization generates the highest revenue?

Visualization:

- Revenue by Specialization

Business Value:

- Analyze department performance

6. Are some doctors overloaded?

Visualization:

- Appointments per Doctor

Business Value:

- Support workforce balancing

Patient Analytics Questions

7. Which age group visits the hospital most frequently?

Visualization:

- Age Distribution

Business Value:

- Improve targeted healthcare services

8. What is the gender distribution of patients?

Visualization:

- Gender Analysis

Business Value:

- Perform demographic analysis

9. Which patients visit repeatedly?

Visualization:

- Repeat Visits

Business Value:

- Monitor chronic care patients

Billing & Revenue Questions

10. Which treatments generate the highest revenue?

Visualization:

- Treatment Revenue

Business Value:

- Perform profitability analysis

11. How many bills are still pending?

Visualization:

- Payment Status

Business Value:

- Monitor revenue recovery

12. Which payment methods are most commonly used?

Visualization:

- Payment Method Analysis

Business Value:

- Analyze customer payment behavior

Operational Questions

13. What is the appointment completion rate?

KPI:

- Completed Appointments / Total Appointments

Business Value:

- Measure operational efficiency

14. Which days or months experience peak hospital traffic?

Visualization:

- Time Trend

Business Value:

- Optimize staffing and resource planning